Multi-Agent Safe Reinforcement Learning-Based Distribution Network Reconfiguration under Cloud-Edge Collaboration

Maoyi Zhou, Zhi Wu, *Senior Member*, *IEEE*, Xiao Zhang, Suyang Zhou, *Senior Member*, *IEEE*, Wei Gu, *Senior Member*, *IEEE*, Ruifeng Zhao

***Abstract*—With the increasing scale of distribution networks and the large-scale integration of distributed energy resources, traditional optimization-based methods encounter scalability limitations in large-scale coupled topology–power flow problems and do not explicitly address tail-risk constraints under high-uncertainty scenarios. This paper proposes a multi-feeder distribution network reconfiguration method based on multi-agent worst-case soft actor-critic safe reinforcement learning within a cloud-edge collaborative framework. Agents are partitioned according to feeders, and a Markov decision process with a globally shared reward is designed under cloud-level centralized training and edge-level decentralized execution. A hard constraint on cumulative safety cost is introduced, while a distributed safety critic network enhances the robustness of reconfiguration strategies in high-risk operating conditions. Simulation results on a practical test system demonstrate that, compared with baseline algorithms, the proposed method reduces voltage deviation by up to 45.87% and decreases network losses by approximately 27.73% under extreme scenarios, while maintaining superior convergence stability.**

***Index Terms*—Large-scale distribution network, distribution network, reconfiguration, cloud-edge collaboration, feeder group, worst-case soft actor-critic algorithm, multi-agent safe reinforcement learning.**

I. Introduction

With the large-scale integration of distributed energy resources (DERs) such as photovoltaic (PV) generation, the traditional operation of distribution networks is undergoing profound transformation [1]. Originally characterized by unidirectional power flow and low complexity, distribution systems are evolving into active distribution networks with features such as multi-source integration, bidirectional power flow, and multi-spatiotemporal coupling [2, 3]. In particular, under the trends of urbanization and increasing electrification, distribution systems exhibit diverse operational control objects and intensified local imbalances, placing higher demands on the collaborative optimization capability of distribution network operation [4, 5].

Against this backdrop, distribution network reconfiguration (DNR) has emerged as a flexible control technology that requires no additional hardware investment and serves as an important means to achieve flexible system operation [6]. By optimizing the network topology, DNR enables rapid response and power flow redistribution in distribution networks, thereby improving operational safety and economic efficiency [7, 8]. Therefore, it is imperative to carry out modeling and solution research for DNR scenarios.

Several scholars have conducted relevant research on DNR. DNR is a large-scale, multi-period coupled, non-convex and mixed-integer optimization problem. Traditional methods mainly describe network topology constraints using mathematical optimization models and solve for optimal or near-optimal DNR schemes via deterministic or heuristic algorithms. Navesi *et al.* [8] proposed a voltage regulation method for distribution networks that integrates switchable capacitor bank configuration with dynamic DNR based on particle swarm optimization. Mokhtari and Gao *et al.* [9, 10] developed a multi-objective DNR model suitable for multi-level switching strategies, combining fuzzy C-means clustering with an improved particle swarm optimization algorithm to effectively alleviate operational issues caused by PV-load imbalances in urban distribution networks. Wang *et al.* [11] addressed the DNR problem with DER integration by proposing an optimization method based on a parallel slime mold algorithm. Through multi-objective modeling and integration with the analytic hierarchy process, this approach significantly improved the efficiency and accuracy of DNR under diverse distributed generation scenarios. Abbas and Li *et al.* [12, 13] proposed a multi-objective bidirectional co-evolutionary algorithm to achieve multi-objective optimization of DER allocation and DNR. However, as the scale of DNR continues to expand, traditional optimization-based methods face challenges in scalability and

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adaptability. To address these limitations, researchers have increasingly turned to data-driven and learning-based approaches to enhance real-time performance and flexibility.

Deep reinforcement learning (DRL) enables "end-to-end" learning and can effectively address the high-dimensional discrete action space optimization challenges in DNR, providing a real-time solution for DNR. Existing studies have made notable progress in algorithm architecture design. Gao *et al.* [4] proposed a two-stage reward-sharing mechanism by quantifying switch contribution, allowing for prioritized control of critical switches. Wang *et al.* [14] combined state serialization modeling with approximate dynamic programming to effectively tackle the curse of dimensionality. Shi *et al.* [15] developed a data-driven batch-constrained algorithm that learns DNR strategies offline from historical data enabling real-time scalability. Bahrami and Hao *et al.* [16, 17] adopted a DRL algorithm based on the transformer architecture and integrated it with neural combinatorial optimization, demonstrating excellent accuracy and scalability in solving DNR problems under load and renewable energy uncertainty. Si *et al.* [18] developed a multi-agent reinforcement learning method for DNR, incorporating a dynamic agent network structure and attention mechanisms to enable flexible perception and coordinated decision-making, while ensuring policy safety through action masking. Gholizadeh *et al.* [19] introduced an explainable DRL decision-making method based on interpreter neural networks, using attribution techniques to reveal the decision logic of DRL models in DNR, thereby enhancing model transparency and trustworthiness. However, existing DNR studies based on DRL primarily design the reward function to emphasize objectives such as network loss reduction and renewable energy accommodation, while paying limited attention to penalizing violations of operational constraints. As the scale of DNR continues to expand, the risk of operational constraint violations becomes increasingly prominent, underscoring the necessity of addressing this issue.

Conventional DRL methods often overlook operational safety constraints, motivating the development of safe DRL frameworks that explicitly account for these constraints and reduce the risk of violations in DNR. Apart from the DNR problem, safe reinforcement learning has found broad applications in power systems, including voltage control [20, 21] and complex power flow computations [22, 23]. However, studies on safe reinforcement learning in the context of DNR remain relatively limited. Gao *et al.* [24] integrated offline and online safe learning mechanisms, then a multi-agent constraint-penalized Q-learning algorithm and a primal-dual multi-agent soft actor-critic (SAC) method were introduced to reduce unsafe trial-and-error exploration and enhance safe policy improvement in large-scale distribution networks. Zhang *et al.* [25] investigated dynamic sequential DNR and a DRL-based approach was proposed to learn sequential switching strategies considering conditional loop and individual switching constraints, effectively preventing partial outages and line overloading in unbalanced distribution systems. Zhan *et al.* [26] developed a bi-level optimization model for DNR using a double deep Q network algorithm embedded with physical knowledge, reducing power losses and voltage deviations while improving computational efficiency.

Recently, to further enhance the practicality and real-time capability of DRL-based DNR methods, cloud–edge collaborative architectures have been proposed, enabling efficient coordination between global and local decision-making processes. In this approach, the cloud platform handles global strategy training and model optimization, while edge devices execute real-time control and decision-making, providing a solution framework for the DNR problem that integrates global information while maintaining the efficiency of distributed decision-making [24, 27]. Gao *et al.* [28] proposed a cloud-edge collaborative framework for DNR effectively reducing photovoltaic curtailment and load shedding and alleviating computational burden in large-scale networks. Chen *et al.* [29] introduced a cloud-edge based approach for urban distribution networks, integrating multi-agent DRL with DNR. The approach supports centralized training and decentralized execution, improving model learning efficiency and enabling real-time DNR. Jia *et al.* [30] optimized edge node configuration through a two-stage design and used credit-based data interaction to ensure reliable coordination of real-time DNR uncertainty.

However, the above studies still suffer from several important limitations. In existing cloud–edge collaboration research, edge agent partitioning is usually restricted to local regions within a single feeder, while agent division in large-scale multi-feeder distribution networks remains largely unexplored. This underscores the need for a physically interpretable agent partitioning strategy tailored to large-scale systems. Moreover, most existing safe DRL methods focus primarily on expected safety costs, with limited consideration of tail risks under extreme conditions, which compromises control reliability in high-risk operating scenarios and calls for further enhancement of safety performance.

To address these research gaps, this paper proposes a cloud–edge collaborative DNR method based on multi-agent worst-case SAC (MA-WCSAC). First, a large-scale DNR model is established with agents partitioned according to feeder groups and physical connections under a cloud-level centralized training and edge-level decentralized execution framework. The DNR problem is then reformulated as a Lagrangian safety-constrained MDP by introducing a hard single-period cumulative safety constraint. Finally, a distributional safety critic replaces the conventional safety critic to enhance risk awareness and control robustness under extreme conditions. Overall, The proposed framework integrates feeder-group-based partitioning, Lagrangian safety constraints and distributional tail-risk modeling within a cloud-edge DNR architecture. Accordingly, this paper makes the following three main contributions:

1. A large-scale DNR model is developed and a MDP is proposed, in which agents are partitioned based on feeder groups and actual physical connections, while adopting a cloud-level centralized training with edge-level decentralized execution framework.

2. The traditional MDP framework is further reformulated as a constrained MDP, and an adaptive safety penalty mechanism is designed based on Lagrangian duality. This mechanism guides the agent to gradually converge 3toward the safe region during training.
3. To enhance the algorithm's safety under extreme uncertainty scenarios, tail risk is modeled by constructing a distributed safety critic that captures the cost distribution in high-risk states.

The rest of this paper is organized as follows. Section II formulates the DNR problem. Section III presents the multi-agent Markov decision process. Section IV proposes the solution algorithm. Section V provides case studies and Section VI concludes the paper.

II. Distribution Network Reconfiguration Model

The DNR framework for the proposed large-scale distribution network is illustrated in Fig. 1. Considering that, in practical distribution networks, feeders are typically organized into feeder groups, and there are usually few or no physical tie lines between different feeder groups, each feeder group generally operates within an independent control area. Based on this characteristic, the agent partitioning in this study is determined according to the feeder-group structure and actual physical connections, rather than by any artificially predefined divisions. These agents are further categorized into multiple distributed edge DNR agents and substation agents. As shown in Fig. 1, the green areas are controlled by DNR agents, each responsible for regulating a set of controllable switches (CS) including tie switches and sectionalizing switches, while the purple areas are supervised by substation agents that primarily control the tap positions of the on-load tap changers (OLTC).

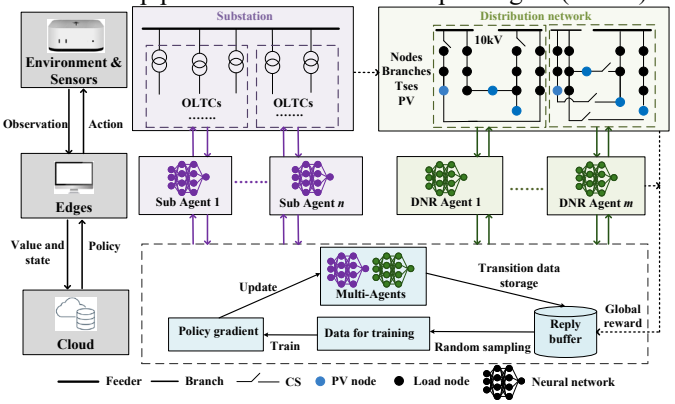


Fig. 1. Framework of DNR under cloud-edge collaboration.

In the edge-level decentralized execution phase, each DNR edge agent collects real-time local node and topology information, while substation edge agents monitor OLTC tap positions and key operational states of connected sub-distribution networks. Based on these observations, agents determine control actions for their assigned equipment and coordinate in real time to achieve near-global optimal operation under distributed decision-making.

In the cloud-level centralized training phase, edge agents upload operational data, state trajectories, and control outcomes to the cloud platform. The cloud aggregates this information to compute a global reward reflecting joint agent strategies, which is stored in the replay buffer and used to update the critic and policy networks via gradient-based learning.

A. Modeling of controllable devices in distribution networks

For volt-var control devices in the distribution network, this paper primarily models PV arrays and OLTCs.

1) Operating constraints of PV arrays

$$0 \le P_t^{\mathrm{PV}} \le P_{\max,t}^{\mathrm{PV}} \tag{1}$$

$$P_{\mathrm{cut},t}^{\mathrm{PV}} = P_{\max,t}^{\mathrm{PV}} - P_t^{\mathrm{PV}} \tag{2}$$

$$P_{\mathrm{cut},t}^{\mathrm{PV}} \le \kappa_{\max}^{\mathrm{PV}} P_{\max,t}^{\mathrm{PV}} \tag{3}$$

$$\sqrt{\left(P_t^{\mathrm{PV}}\right)^2 + \left(Q_t^{\mathrm{PV}}\right)^2} \le S^{\mathrm{PV}} \tag{4}$$

where $P_{\max,t}^{\mathrm{PV}}$ represents the maximum PV output at time t; P_t^{PV} denotes the actual PV output at time t; Q_t^{PV} is the reactive power absorbed or generated by the PV system at time t; $P_{\mathrm{cut},t}^{\mathrm{PV}}$ indicates the curtailed PV power at time t; $\kappa_{\max}^{\mathrm{PV}}$ is the upper limit of the PV curtailment rate [31]; S^{PV} refers to the installed capacity of the PV arrays.

2) Operating constraints of OLTCs

An OLTC is installed between the upstream grid and the root node of the distribution network. Let $U_{i,t}^{\mathrm{up}}$ denote the voltage of the upstream grid. Then, the squared voltage at the root node of the distribution network can be expressed as:

$$\left(U_{i,t}\right)^2 = \left(1 + \iota_{ij,t}\Delta U\right)^2 \cdot \left(U_{i,t}^{\mathrm{up}}\right)^2, \forall (ij) \in \Omega_{\mathrm{OL}} \tag{5}$$

where Ω_{OL} denotes the set of branches equipped with OLTCs; ij denotes the index of the branch between nodes i and j; $\iota_{ij,t}$ represents the center tap position of the OLTC; ΔU is the per-unit voltage value adjustable by a single tap position.

B. Radial topology constraints of the distribution network

Radial topology constraints are one of the essential constraints that must be satisfied during DNR [32]. Specifically, the switch status combinations of tie-line switches in the distribution network need to satisfy the following two conditions: (1) At least one branch in each fundamental cycle must be open to avoid the formation of loops; (2) Among the common branches shared by two fundamental cycles, no more than one branch is allowed to be closed to prevent the occurrence of islands:

$$\mu_{ij,t}^{\mathrm{CS}} \in \{0,1\}, \forall (ij) \in \Omega_{\mathrm{CS}} \tag{6}$$

$$n_Z - \left(n_J - n_B\right) = \sum\nolimits_{(ij)\in\Omega_{\mathrm{CS}}} \left(1 - \mu_{ij,t}^{\mathrm{CS}}\right) \tag{7}$$

$$\sum_{(ij)\in\Omega_{\mathrm{CS}}} B_{m,(ij)} \mu_{ij,t}^{\mathrm{CS}} \ge 1, \quad \forall m \in \Omega_{\mathrm{Bus}} \tag{8}$$

where Ω_{CS} denotes the set of branches equipped with CSs; Ω_{Bus} denotes the set of nodes; n_Z, n_J and n_B represent the number of branches, nodes and feeder root nodes, respectively; $\mu_{ij,t}^{\mathrm{CS}}$ is a set of binary variables indicating the switching status of CSs at time t. In this paper, $\mu_{ij,t}^{\mathrm{CS}} = 0$ denotes an open CS, while $\mu_{ij,t}^{\mathrm{CS}} = 1$ indicates a closed CS; $B \in \{0,1\}^{n_J \times n_Z}$ is the node-branch incidence matrix, whose element $B_{m,(ij)} = 1$ if branch (i, j) is incident to node m, and $B_{m,(ij)} = 0$ otherwise.

C. Power flow constraints of the distribution network

Under condition $(ij) \notin \Omega_{\mathrm{OL}}$, the following constraints apply:

$$\left(U_{i,t}\right)^2 \cdot \left(I_{ij,t}\right)^2 \geq \left(P_{ij,t}\right)^2 + \left(Q_{ij,t}\right)^2 \tag{9}$$

$$\sum_{i\in\Omega_m(j)} \left[P_{ij,t} - r_{ij}\left(I_{ij,t}\right)^2\right] = \sum_{k\in\Omega_n(j)} P_{jk,t} - \left(-P_{j,t}^{\mathrm{load}} + P_{j,t}^{\mathrm{PV}}\right) \tag{10}$$

$$\sum_{i\in\Omega_m(j)} \left[Q_{ij,t} - x_{ij}\left(I_{ij,t}\right)^2\right] = \sum_{k\in\Omega_n(j)} Q_{jk,t} - \left(-Q_{j,t}^{\mathrm{load}} + Q_{j,t}^{\mathrm{PV}}\right) \tag{11}$$

$$\mu_{ij,t}^{\mathrm{BR}} = \begin{cases} \mu_{ij,t}^{\mathrm{CS}}, \forall (ij) \in \Omega_{\mathrm{CS}} \\ 1, \text{else} \end{cases} \tag{12}$$

$$U_{j,t}^2 - U_{i,t}^2 \leq M(1-\mu_{ij,t}^{\mathrm{BR}}) - 2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) + (r_{ij}^2 + x_{ij}^2)I_{ij,t}^2 \tag{13}$$

$$U_{j,t}^2 - U_{i,t}^2 \geq M(\mu_{ij,t}^{\mathrm{BR}} - 1) - 2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) + (r_{ij}^2 + x_{ij}^2)I_{ij,t}^2 \tag{14}$$

$$U_{j,\min} \leq U_{j,t} \leq U_{j,\max} \tag{15}$$

$$I_{ij,t} \leq \mu_{ij,t}^{\mathrm{BR}} I_{ij,\max} \tag{16}$$

where $\Omega_m(j)$ denotes the set of sending-end nodes of branches that terminate at node j in the distribution network; $\Omega_n(j)$ denotes the set of receiving-end nodes of branches that originate from node j in the distribution network; $U_{j,t}$ denotes the voltage magnitude at node j at time t; $P_{ij,t}$ and $Q_{ij,t}$ represent the active and reactive power at the sending end of branch ij at time t respectively; $P_{j,t}^{\mathrm{load}}$ and $Q_{j,t}^{\mathrm{load}}$ are the active and reactive power demands of the load at time t respectively; r_{ij} and x_{ij} denote the resistance and reactance of branch ij respectively; M is a sufficiently large constant [33]; $U_{j,\max}$ and $U_{j,\min}$ represent the lower and upper voltage limits of node j; $I_{ij,\max}$ is the current upper limit of branch ij; $\mu_{ij,t}^{\mathrm{BR}}$ indicates the switch status of the branch.

Under condition $(ij) \in \Omega_{\mathrm{OL}}$, the branch equipped with an OLTC is modeled as follows:

$$P_{ij,t} - r_{ij}\left(I_{ij,t}\right)^2 + P_{i,t} = \sum_{k\in\Omega_n(j)} P_{jk,t} \tag{17}$$

$$Q_{ij,t} - x_{ij}\left(I_{ij,t}\right)^2 + Q_{i,t} = \sum_{k\in\Omega_n(j)} Q_{jk,t} \tag{18}$$

$$U_{i,t}{}^2 - U_{m,t}{}^2 = 2\left(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}\right) - \left(r_{ij}{}^2 + x_{ij}{}^2\right)I_{ij,t}{}^2 \tag{19}$$

$$\left(U_{i,t}\right)^2 \cdot \left(I_{ij,t}\right)^2 = \left(P_{ij,t}\right)^2 + \left(Q_{ij,t}\right)^2 \tag{20}$$

$$\left(U_{j,t}\right)^2 = \left(1 + \iota_{ij,t}\Delta U\right)^2 \cdot \left(U_{m,t}\right)^2 \tag{21}$$

D. Objective function

The objective function (22) is constructed with different weights to comprehensively consider these factors:

$$\begin{cases} \min \sum_{t=1}^{T} \left(\begin{array}{l} c_1 \sum_{i=1}^{n_J} (U_{i,t} - 1)^2 + c_2 \sum_{i=1}^{n_J} \sum_{j\in\Omega_n(i)} I_{ij,t}{}^2 r_{ij} \Delta t \\ + c_3 \sum_{(ij)\in\Omega_{\mathrm{TS}}} \left|\mu_{ij,t+1}^{\mathrm{CS}} - \mu_{ij,t}^{\mathrm{CS}}\right| + c_4 \sum_{(ij)\in\Omega_{\mathrm{OL}}} \left|\iota_{ij,t+1} - \iota_{ij,t}\right| \\ + c_5 \sum_{i\in\Omega_{\mathrm{Sub}}} (P_{i,t} - \overline{P}_t)^2 \end{array} \right) \\ s.t. \text{Equations (1)-(21)} \end{cases} \tag{22}$$

where $\overline{P}$ is the active power transmitted by the feeders; $P_{i,t}, i \in \Omega_{\mathrm{Sub}}$ is the average power flow across all feeders.

The terms in equation (22) represent voltage deviation, network losses, CS switching cost, OLTC switching cost and feeder load balancing respectively [34, 35].

III. Multi-Agent Distribution Network Reconfiguration Markov Decision Process

For the large-scale multi-feeder distribution network, the system is first divided into two main zones: the distribution network and the substation group. Then, based on feeder group division and the actual physical connections, each of these main zones can be further subdivided into local zones. This hierarchical partitioning helps reduce the control complexity for individual agents and also provides a more intuitive physical interpretation.

Clearly, the above model requires the introduction of multi-agent joint optimization learning. The multi-agent deep reinforcement learning process can also be described by a MDP [36]. Such an MDP can be represented as a five-tuple $(\boldsymbol{S}, \boldsymbol{A}, \boldsymbol{P}, \boldsymbol{R}, \gamma)$. In this paper, two groups of agents, Ψ_{DN} and Ψ_{Sub}, are defined to respectively control sub-distribution network DNR and substation tap position selection. Let these two groups consist of n_{DN} and n_{Sub} agents. γ is the discount factor used to adjust the optimization objective.

A. State space

The observation state of each agent in this paper is represented as a vector composed of physical quantities at each time step. Specifically, the observation state of sub-distribution network DNR agents consists of the node voltages, active and reactive load power, active and reactive PV output and a one-hot vector representing the system topology within the agent's controlled zone. The one-hot vector for system topology indicates the open/closed status of all branches within the zone, where 0 denotes a closed branch and 1 denotes an open branch. The observation state of a substation agent is composed of the OLTC tap position and the observation states of the adjacent sub-distribution network zones on either side. The detailed description of the state space is given in equations (23)–(24):

$$s_t = \left\{s_{t,g_1}^{\mathrm{DN}}, s_{t,g_2}^{\mathrm{Sub}}\right\} \in \boldsymbol{S},\ g_1 \in \Psi_{\mathrm{DN}}, g_2 \in \Psi_{\mathrm{Sub}} \tag{23}$$

$$\begin{cases} s_{t,g_1}^{\mathrm{DN}} = \left\{U_{t,g_1(j_1)}, P_{t,g_1(j_1)}^{\mathrm{load}}, Q_{t,g_1(j_1)}^{\mathrm{load}}, P_{t,g_1(j_1)}^{\mathrm{PV}}, \xi_{t,g_1(j_1)}\right\}, j_1 \in [1, n_{\mathrm{DN}}] \\ s_{t,g_2}^{\mathrm{Sub}} = \left\{s_{t,g2(j2)}^{\mathrm{DN}}\right\} \bigcup \left\{\iota_{t,g_2(j_2)}\right\}, j_2 \in [1, n_{\mathrm{Sub}}] \end{cases} \tag{24}$$

where s_{t,g_1}^{DN} and s_{t,g_2}^{Sub} represent the observation states at time t for the entire distribution network and the entire substation group respectively; $s_{t,g2(j_2)}^{\mathrm{DN}}$ represent observation states of the sub-distribution network zones that are adjacent to the substation region controlled by the j_2-th agent in Ψ_{Sub} at time t; $U_{t,g_1(j_1)}$ represents the node voltage components at time t within the sub-distribution network zone controlled by the j_1-th agent in Ψ_{DN}; $\xi_{t,g_1(j_1)}$ is a one-hot vector representing the topology

of the region controlled by the corresponding agent. Other physical quantities are defined similarly.

B. Action space

This chapter continues to focus on investigating the coordination performance between DNR and OLTCs. The action space is defined as follows:

$$a_t = \left\{ a^{\mathrm{DN}}_{t,g_1}, a^{\mathrm{Sub}}_{t,g2} \right\} \in \boldsymbol{A},\ g_1 \in \Psi_{\mathrm{DN}}, g_2 \in \Psi_{\mathrm{Sub}} \tag{25}$$

$$\begin{cases} a^{\mathrm{DN}}_{t,g_1} = \left\{ a^{\mathrm{CS}}_{t,g_1(j_1)} \right\}, j_1 \in [1, n_{\mathrm{DN}}] \\ a^{\mathrm{Sub}}_{t,g2} = \left\{ \iota^{\mathrm{OLTC}}_{t,g_2(j_2)} \right\}, j_2 \in [1, n_{\mathrm{Sub}}] \end{cases} \tag{26}$$

where a^{DN}_{t,g_1} and $a^{\mathrm{Sub}}_{t,g2}$ represent the action spaces at time t for the entire distribution network and the entire substation group; $a^{\mathrm{CS}}_{t,g_1(j_1)}$ denotes the set of DNR schemes for the sub-distribution network zone controlled by the j_1-th agent in Ψ_{DN}.

$\iota^{\mathrm{OLTC}}_{t,g_2(j_2)}$ denotes the set of OLTC tap position adjustment actions at time t for the substation region controlled by the j_2-th agent in Ψ_{Sub}. In this paper, it is assumed that $-5 \le \iota_t \le 5$, i.e., each OLTC has 11 adjustable tap positions and $\Delta U = 0.01$ p.u., so the size of $a^{\mathrm{Sub}}_{t,g2}$ is $11^{N_{\mathrm{OLTC}}}$ and N_{OLTC} represents the total number of OLTCs in the entire grid.

The action space of DNR are determined by 0–1 variables, and it is evident that their size grows exponentially with the number of CSs. Reference [32] proposes a DFS-based space reduction method to filter solutions satisfying radial topology constraints. However, it still suffers from an excessively large action space in large-scale networks. To address this problem, this paper adopts the "switch pair" method to further reduce the action space [17]. Specifically, the "switch pair" method allows: (1) switch operations to be limited to paired actions, i.e., a combination of one opening operation and one closing operation; (2) maintaining the current network topology, i.e., no switching action.

The action space of a DNR agent based on the "switch pair" method along with its size, is defined in equations (27)–(28):

$$a^{\mathrm{DN}}_{t,g} = \left\{ (i,j) \middle| i,j \in \Omega^{\mathrm{DN}}_{g,\mathrm{TS}}, i \ne j \right\} \cup \left\{ (0,0) \right\}, g \in \left\{ \Psi_{\mathrm{DN}} \right\} \tag{27}$$

$$\left| a^{\mathrm{DN}}_{t,g} \right| = \left(n^{\mathrm{DN}}_{g,\mathrm{TS}} \right)^2 - n^{\mathrm{DN}}_{g,\mathrm{TS}} + 1, g \in \left\{ \Psi_{\mathrm{DN}} \right\} \tag{28}$$

where $n^{\mathrm{DN}}_{g,\mathrm{TS}}$ denote the number of CSs within the control region of DNR agent g.

Equation (27) defines the operation rules for a switch pair (i, j) and (0, 0) indicates that all CSs remain in their original states.

It should be noted that PV active power is not directly included in the agent action space. Instead, it is determined by solving the lower-level quadratically constrained programming problem, which ensures power flow feasibility and operational constraints. The RL agents primarily determine topology and discrete control actions, while continuous variables such as PV output are computed within the optimization layer.

C. State transition

The state transition process is defined as the state transition function $s_{t+1} = P(s_t, a_t), \boldsymbol{P}: \boldsymbol{S} \times \boldsymbol{A} \to \boldsymbol{S}$, which represents the transition from a current state s_t to a subsequent state s_{t+1}. In terms of the time scale, this paper adopts a discrete time step of $\Delta t = 1$ hour to perform the state transitions [16, 17].

D. Reward function

Each agent in this paper is assigned a shared reward signal. The reward function (29) is further modified based on (22):

$$r_t = \begin{pmatrix} \sum_{i=1}^{n_J} \left(-c_1 (U_{i,t} - 1)^2 - c_2 \sum_{j \in \Omega_n(i)} I_{ij,t}{}^2 r_{ij} \Delta t \right) \\ -c_3 \sum_{(ij) \in \Omega_{\mathrm{TS}}} \left| \mu^{\mathrm{CS}}_{ij,t+1} - \mu^{\mathrm{CS}}_{ij,t} \right| - c_4 \sum_{(ij) \in \Omega_{\mathrm{OL}}} \left| \iota_{ij,t+1} - \iota_{ij,t} \right| \\ -c_5 \sum_{i \in \Omega_{\mathrm{Sub}}} (P_{i,t} - \bar{P}_t)^2 - \Im_{\mathrm{end}} \end{pmatrix} \tag{29}$$

Compared to equation (22), the reward function introduces an abnormal operation penalty term $\Im_{\mathrm{end}}$. The following clarifications are provided regarding $\Im_{\mathrm{end}}$: (1) When the agent completes the predefined training episode T and transitions to the next episode, this is considered a normal learning process and the penalty term $\Im_{\mathrm{end}}$ is not triggered; (2) When $|U_{i,t} - 1|$ exceeds 0.05 p.u. or numerical failures occur—such as power flow calculation not converging—the penalty term $\Im_{\mathrm{end}}$ is activated in the reward function to ensure that the agent becomes aware of the severity of the abnormal operating condition caused by its actions.

IV. Solution Algorithm

A. Safe mechanism based on constrained Markov decision process and Lagrangian duality

SAC has demonstrated strong performance and stability in continuous control tasks. However, it lacks explicit modeling of safety constraints, which are critical in power system operations. To address this, this paper introduces the MA-WCSAC algorithm, whose framework is showed in Fig. 2.

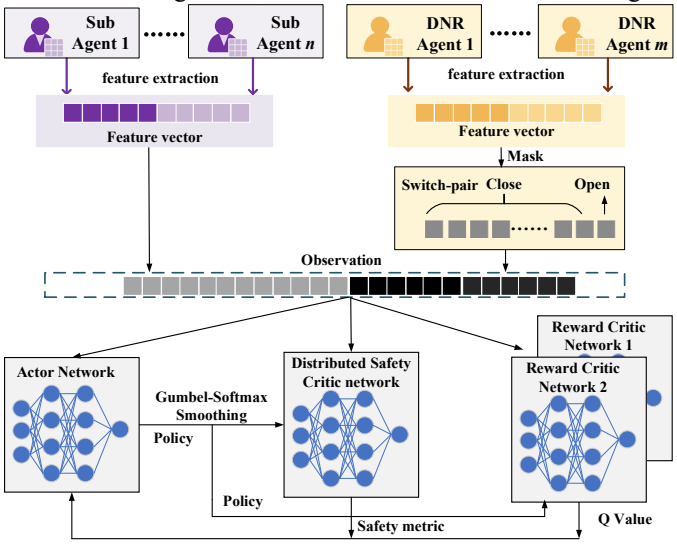


Fig. 2. Framework of MA-WCSAC algorithm.

The cloud layer performs global policy optimization through the reward and safety critic networks, while edge agents

execute local control via actor networks with embedded safety modeling. Taking DNR agents as an example, a mask generated by the "switch pair" method is combined with network physical information and fed into the actor, reward critic and distributional safety critic networks. The actor outputs a policy which is evaluated by the two critics to produce the Q-value and safety metric for policy updates.

In Section III, the original DNR problem is modeled as a standard MDP defined by $(\boldsymbol{S},\boldsymbol{A},\boldsymbol{P},\boldsymbol{R},\gamma)$. However, a standard MDP does not explicitly consider safety constraints. To align with the MA-WCSAC algorithm proposed below in this paper, the MDP is reformulated as a constrained MDP (CMDP), which is described by a seven-tuple $(\boldsymbol{S},\boldsymbol{A},\boldsymbol{P},\boldsymbol{R},\boldsymbol{\upsilon},\zeta,\gamma)$ [37], where two new elements $\boldsymbol{\upsilon}$ and ζ are added. $\boldsymbol{\upsilon}$ denotes the safety cost associated with the state-action pair (s, a), which quantifies the penalty incurred when operational constraints are violated. ζ is the safety threshold, specifying the maximum allowable cumulative safety cost.

Let θ_g parameterize the actor network. A Lagrange multiplier λ_g is introduced to construct a DRL optimization model with safety constraints:

$$\begin{aligned}\min_{\lambda\geq 0}\max_{\pi_g} L=&\sum_{t=1}^{T}E_{s_g,a_g\sim\rho_{\pi_{\theta_g}}}\left[\gamma^t r_{t,g}\left(s_{t,g},a_{t,g}\right)+\alpha_g H\left(\pi_{\theta_g}\right)\right]\\&-\lambda_g\left(\sum_{t=1}^{T}E_{s_g,a_g\sim\rho_{\pi_{\theta_g}}}\left[\gamma^t \upsilon_{t,g}\left(s_{t,g},a_{t,g}\right)\right]-\zeta_g\right)\end{aligned}\tag{30}$$

where π_{θ_g} is the policy; $\rho_{\pi_{\theta_g}}$ is the policy trajectory of agent g; ζ_g is the upper bound of the expected long-term safety cost for agent g; α_g is the entropy regularization coefficient for agent g; $\upsilon_{t,g}$ is the cost function which equals 1 if taking action $a_{t,g}$ in state $s_{t,g}$ violates a safety constraint and 0 otherwise.

B. Worst-case mechanism based on distributional safety critic and conditional value-at-risk

For the MA-WCSAC algorithm, two types of critic networks are required to adaptively balance reward and safety [38]. The reward critic network is defined to estimate the long-term reward. The safety critic network is defined to estimate the cumulative safety cost, ensuring the safety of the policy.

The safety critic network is defined to estimate the expected long-term cumulative cost $Q^c_{\pi_{\theta_g}}$ starting from the initial state, where the cumulative cost is denoted as $C^c_{\pi_{\theta_g}}$. In safety-critical domains such as power system operation, the optimal policy must not only aim to maximize rewards but also maintain robustness in the face of uncertainty and risk. However, long-term cumulative cost evaluates only the expected value of the safety cost. This expectation reflects the average level of safety cost but it fails to account for variability and tail risks. Therefore, this paper further adopts a distributional safety critic network to replace the safety critic network.

When following policy π_{θ_g} and the distribution $P\left[C^c_{\pi_{\theta_g}}\middle|s_{t,g},a_{t,g}\right]$ of the long-term safety cost, the Bellman operator ξ for the distributional safety critic network is defined as follows:

$$\xi C^c_{\pi_{\theta_g}}\left(s_{t,g},a_{t,g}\right)=\upsilon_{t,g}\left(s_{t,g},a_{t,g}\right)+\gamma C^c_{\pi_{\theta_g}}\left(s_{t+1,g},a_{t+1,g}\right)\tag{31}$$

$$C^c_{\pi_{\theta_g}}\left(s_{t,g},a_{t,g}\right)\sim\mathcal{N}\left(Q^c_{\pi_{\theta_g}}\left(s_{t,g},a_{t,g}\right),V^c_{\pi_{\theta_g}}\left(s_{t,g},a_{t,g}\right)\right)\tag{32}$$

This paper employs two neural networks parameterized by μ_g and η_g respectively to evaluate the distributional safety critic network. Specifically, μ_g and η_g parameterize the mean and variance of the distribution of the distributional safety critic network respectively:

$$\begin{cases}Q^c_{\pi_{\theta_g}}\left(s_{t,g},a_{t,g}\right)\leftarrow Q^c_{\mu_g}\left(s_{t,g},a_{t,g}\right)\\V^c_{\pi_{\theta_g}}\left(s_{t,g},a_{t,g}\right)\leftarrow V^c_{\eta_g}\left(s_{t,g},a_{t,g}\right)\end{cases}\tag{33}$$

Among various risk metrics, the conditional value-at-risk (CVaR) is adopted to quantify the worst-case tail expectation, providing a rigorous safety guarantee. As training progresses, π_{θ_g} and $P\left[C^c_{\pi_{\theta_g}}\middle|s_{t,g},a_{t,g}\right]$ dynamically change. As long as the tail of $P\left[C^c_{\pi_{\theta_g}}\middle|s_{t,g},a_{t,g}\right]$, specifically, the pre-defined $CVaR$ value is maintained below a certain threshold, the resulting policy can be considered safe.

Let the cumulative distribution function of $F_{C,g}$ be $P\left[C^c_{\pi_{\theta_g}}\middle|s_{t,g},a_{t,g}\right]$ and let $F^{-1}_{C,g}$ be the inverse function of $F_{C,g}$. Then $CVaR_{\varpi_g}$ is defined as follows:

$$CVaR_{\varpi_g}=E_P\left[C^c_{\pi_{\theta_g}}\middle|C^c_{\pi_{\theta_g}}\geq F^{-1}_{C,g}\left(1-\varpi_g\right)\right]\tag{34}$$

where the coefficient $\varpi_g\in(0,1]$ is used to define the safety risk level in the WCSAC algorithm. The closer ϖ_g is to 0, the more the agent emphasizes risk avoidance. Conversely, as ϖ_g approaches 1, the agent places more emphasis on obtaining higher reward values. $F^{-1}_{C,g}\left(1-\varpi_g\right)$ denotes the value of the long-term safety cost function when the cumulative probability reaches $1-\varpi_g$.

$CVaR_{\varpi_g}$ is calculated by evaluating $Q^c_{\mu_g}\left(s_{t,g},a_{t,g}\right)$ and $V^c_{\eta_g}\left(s_{t,g},a_{t,g}\right)$ in equation (33):

$$\begin{aligned}CVaR_{\varpi_g}&=\chi_{\pi_{\theta_g}}\left(s_{t,g},a_{t,g},\varpi_g\right)\\&=Q^c_{\pi_{\theta_g}}\left(s_{t,g},a_{t,g}\right)+\varpi_g{}^{-1}\phi\left(\Phi^{-1}\left(\varpi_g\right)\right)\sqrt{V^c_{\pi_{\theta_g}}\left(s_{t,g},a_{t,g}\right)}\end{aligned}\tag{35}$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ represent the probability density function and cumulative distribution function of the Gaussian distribution $\mathcal{N}$ respectively.

For a predefined ϖ_g and any time step, the policy π_{θ_g} must be optimized until it satisfies equation (36):

$$\chi_{\pi_{\theta_g}}\left(s_{t,g},a_{t,g},\varpi_g\right)\le\zeta_g \tag{36}$$

C. Action sampling and masking for discrete distribution network reconfiguration control

It is evident that the action spaces of DNR and OLTC are discrete. To address the non-differentiability of discrete actions in SAC, this paper adopts the Gumbel-Softmax reparameterization technique [17], which approximates discrete distributions with continuous ones, enabling gradient backpropagation for policy optimization:

$$\bar{\pi}_{\theta_g}=\mathrm{softmax}\left(\left(f_{\theta_g}\left(s_{t,g}\right)+G_n\right)/\tau_{te}+\log\Lambda_{t,g}\right) \tag{37}$$

$$G_n=-\log\left(-\log u\right),u\sim\mathrm{Uniform}\left(0,1\right) \tag{38}$$

where G_n is noise following a Gumbel distribution; τ_{te} is the temperature coefficient; $\bar{\pi}_{\theta_g}$ is the sampled policy; $\Lambda_{t,g}$ is a binary matrix used to filter out invalid switch-pair actions which can be obtained according to [20].

At this point, $s_{t,g}$ and $a_{t,g}$ are approximated by $\bar{\pi}_{\theta_g}$ and $\hat{a}_{t,g}$. Let $\wp_{\bar{\pi}_{\theta_g}}=Q^r_{\bar{\pi}_{\theta_g}}\left(s_{t,g},\hat{a}_{t,g}\right)-\lambda_g\chi_{\bar{\pi}_{\theta_g}}\left(s_{t,g},\hat{a}_{t,g},\varpi_g\right)$, then the loss function of the actor network is given by:

$$J_{\bar{\pi}}\left(\theta_g\right)=E_{s_g\sim D,\hat{a}_g\sim\rho_{\bar{\pi}_{\theta_g}}}\left[\alpha_g\log_{\bar{\pi}_{\theta_g}}\left(\hat{a}_{t,g}\middle|s_{t,g}\right)-\wp_{\bar{\pi}_{\theta_g}}\left(s_{t,g},\hat{a}_{t,g}\right)\right] \tag{39}$$

The loss function of λ_g can be obtained as follows:

$$J_{\lambda_g}\left(\lambda_g\right)=E_{s_g\sim D,\hat{a}_g\sim\rho_{\bar{\pi}_{\theta_g}}}\left[\lambda_g\left(\zeta_g-\chi_{\bar{\pi}_{\theta_g}}\left(s_{t,g},\hat{a}_{t,g},\varpi_g\right)\right)\right] \tag{40}$$

The update rules for μ_g and η_g of the distributional safety critic network are as follows:

$$J_\mu\left(\mu_g\right)=E_{\left(s_{t,g},\hat{a}_{t,g}\right)\sim D}\left\|\Delta Q\left(s_{t,g},\hat{a}_{t,g},\mu_g\right)\right\|_2^2 \tag{41}$$

$$J_\eta\left(\eta_g\right)=E_{\left(s_{t,g},\hat{a}_{t,g}\right)\sim D}trace\left(\Delta V\left(s_{t,g},\hat{a}_{t,g},\eta_g\right)\right) \tag{42}$$

where $trace(\bullet)$ is the trace of a matrix; $\|\cdot\|_2^2$ refers to the squared L_2-norm; J_μ is the error function of $Q^c_{\mu_g}\left(s_{t,g},a_{t,g}\right)$; J_η is the error function of $V^c_{\eta_g}\left(s_{t,g},a_{t,g}\right)$.

Two reward critic networks are parameterized by $\psi_{w,g},w\in\{1,2\}$ and two reward target-critic networks are parameterized by $\bar{\psi}_{w,g},w\in\{1,2\}$. Similarly, two distributional safety target-critic networks are also defined and parameterized by $\bar{\mu}_g$ and $\bar{\eta}_g$ respectively. The updates of the reward critic network are shown in equations (43)–(44):

$$\nabla_{\psi_{w,g}}J_w(\psi_{w,g})=\nabla_{\psi_{w,g}}\left\|y_{t,g}-Q^r_{\psi_{w,g}}(s_{t,g},\hat{a}_{t,g})\right\|^2 \tag{43}$$

$$y_{t,g}=r_{t,g}+\gamma\min_{w=1,2}(Q^r_{\bar{\psi}_{w,g}}(s_{t+1,g},\hat{a}_{t+1,g}))+\alpha_g{}^TH(\bar{\pi}_{\theta_g}) \tag{44}$$

The update methods for reward target-critic networks and distributional safety target-critic networks are shown in equations (45)–(47):

$$\bar{\psi}_{w,g}=\delta_{\mathrm{I},w,g}\psi_{w,g}+\left(1-\delta_{\mathrm{I},w,g}\right)\bar{\psi}_{w,g},w\in\{1,2\} \tag{45}$$

$$\bar{\mu}_g=\delta_{\mathrm{II},g}\mu_g+\left(1-\delta_{\mathrm{II},g}\right)\bar{\mu}_g \tag{46}$$

$$\bar{\eta}_g=\delta_{\mathrm{III},g}\eta_g+\left(1-\delta_{\mathrm{III},g}\right)\bar{\eta}_g \tag{47}$$

where $\delta_{\mathrm{I},w,g}$, $\delta_{\mathrm{II},g}$ and $\delta_{\mathrm{III},g}$ are weighting coefficients.

D. Solution procedure

The solution procedure of the MA-WCSAC algorithm is detailed in Algorithm 1. In this study, the convergence of MA-WCSAC is determined based on both training stability and safety satisfaction. Specifically, training is conducted for a predefined number of episodes, and convergence is assessed by observing the stabilization of the reward curve. Meanwhile, the learned policy is required to satisfy the predefined CVaR-based safety threshold. The algorithm is considered converged once both performance stability and safety constraints are met, at which point training proceeds until the predefined number of episodes is reached.

Algorithm 1: DNR Method Based on MA-WCSAC

Initialize the parameters θ_g, $\psi_{w,g}$, μ_g, η_g, λ_g
Initialize the parameters $\langle\bar{\psi}_{w,g},\bar{\mu}_g,\bar{\eta}_g\rangle\leftarrow\langle\psi_{w,g},\mu_g,\eta_g\rangle$
Initialize the experience replay buffer D
Set the counter $NUM = 0$
For episode = 1 : N_e **do**
  Initialize the agent's observation environment
  For t = 1 : T **do**
    Select DNR and OLTC regulation schemes from $\boldsymbol{A}$
    Call (29) to obtain $r_{t,g}$
    Store the experience $\{s_{t,g},a_{t,g},r_{t,g},s_{t+1,g}\}$ into D
    $NUM = NUM + 1$
    If $NUM > NUM_{ex}$ **then**
      Call (39) to update θ_g
      Call (40) to update λ_g
      Call (41)-(42) to update μ_g, η_g
      Call (43)-(44) to update $\psi_{w,g}$
      Call (45) to update $\bar{\psi}_{w,g}$
      Call (46)-(47) to update $\bar{\mu}_g$, $\bar{\eta}_g$
    End if
    Update the state $s_{t,g}$ to $s_{t+1,g}$
  End for
End for

V. Case Study

To verify the effectiveness, superiority, and safety of the proposed algorithm, we select a real distribution network as shown in Fig. 3 as the case study. A total of five agents are configured. Specifically, Agent 1 is responsible for controlling nodes 4–59. Two agents manage the substation OLTCs. Agents 4 and 5 control nodes 60–159 and 160–289, respectively. The agents are partitioned according to the actual physical connections and feeder-group structure, and the superiority of this partitioning strategy is verified in Section V.D. The allowable upper and lower voltage limits are 1.05 p.u. and 0.95 p.u. respectively. The parameters of the sub-distribution

networks are listed in TABLE I. The PV output and load variation curves over one operation cycle are shown in Fig. 4.

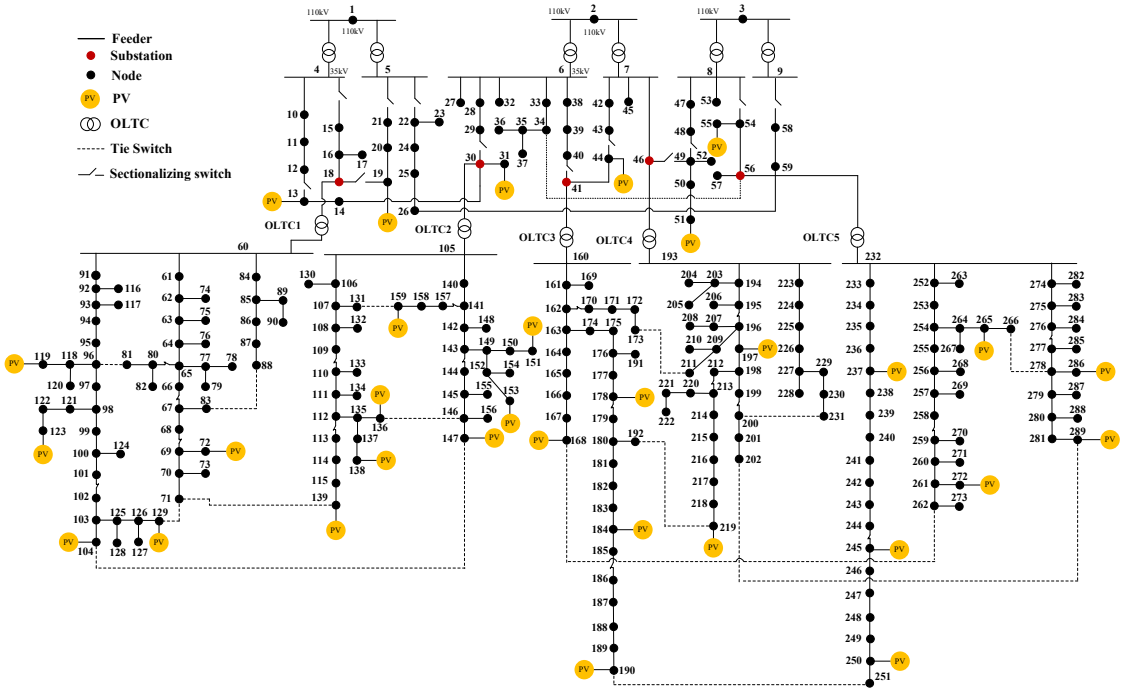


Fig. 3. Technical parameters of the 289-node power system.

TABLE I
TECHNICAL PARAMETER OF THE 289-NODE POWER SYSTEM

Parameter	HV Agent 1	MV Agent 4	MV Agent 5
CS	13	15	16
Branch	56	105	134
Feeder	6	2	3
Bus	56	100	130
PV	6	12	13

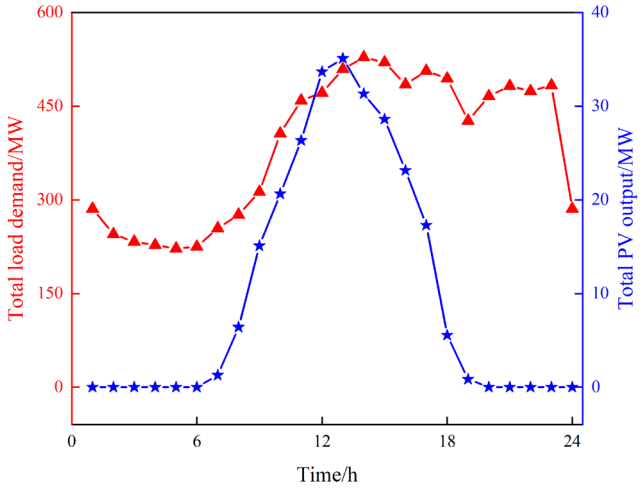


Fig. 4. Curve of PV output and load.

In the DRL algorithm, the evaluation networks consist of one input layer, two hidden layers and one output layer, with ReLU functions used as activation functions. The algorithm is implemented based on PyTorch under Python 3.9. The simulation environment for the case study includes an Intel(R) Core(TM) i9-13900HX 2.20GHz CPU, 32GB of RAM and an RTX 4070 Laptop GPU. The algorithm parameters are listed in TABLE II, in which the term "uniform" in parentheses indicates that the value is taken in the same way for each agent.

TABLE II
ALGORITHM TRAINING PARAMETER SETTINGS

Parameter	Value (Agent 1/2/3/4/5)
Input layer neurons	280/788/938/505/654
Hidden layer 1 neurons	1024 (uniform)
Hidden layer 2 neurons	1024 (uniform)
Replay buffer size	3000 (uniform)
T	24 (uniform)
γ	0.99 (uniform)
τ_{te}	1.2 (uniform)
α_g (actor/critic/target-critic)	0.0001/0.0003/0.0003
ϖ_g	0.5 (uniform)
ζ_g	1 (uniform)
Batch size	64
Size of D	10000
$c_1/c_2/c_3/c_4/c_5$	0.5/0.4/0.25/0.9/0.25
Training episodes	600 (uniform)
Optimizer	Adam (uniform)

A. *Convergence performance of algorithms*

This subsection compares the convergence performance of the proposed MA-WCSAC algorithm with the MASAC (multi-agent SAC) and MAPPO (multi-agent proximal policy optimization) algorithms. MAPPO is an on-policy multi-agent extension of PPO under centralized training and decentralized execution, widely used as a benchmark in cooperative control tasks. The reward curves during the training phases are shown in Fig. 5. It should be noted that the legends "-0.1", "-0.5" and "-0.9" correspond to the cases where the parameter ϖ_g in the MA-WCSAC algorithm is set to 0.1, 0.5 and 0.9, respectively. To ensure the smoothness of the curves, a moving average smoothing with a window size of 15 is applied.

The MASAC algorithm (In this paragraph, we only discuss the case where $\varpi_g = 0.5$) converges rapidly, typically within about 70 training episodes, but it is prone to premature

convergence to local optima. In contrast, MA-WCSAC incorporates state transitions under extreme scenarios during policy optimization, leading to slower convergence but improved overall performance. Specifically, MA-WCSAC extends MASAC by introducing a hard safety constraint and a distributional safety critic network. The constraint smooths the policy gradient, while the risk-aware signals from the distributional safety critic mitigate value overestimation. Although these additions increase the number of training parameters and convergence complexity, Fig. 5 shows that MA-WCSAC achieves superior convergence stability and control performance compared with the other two algorithms.

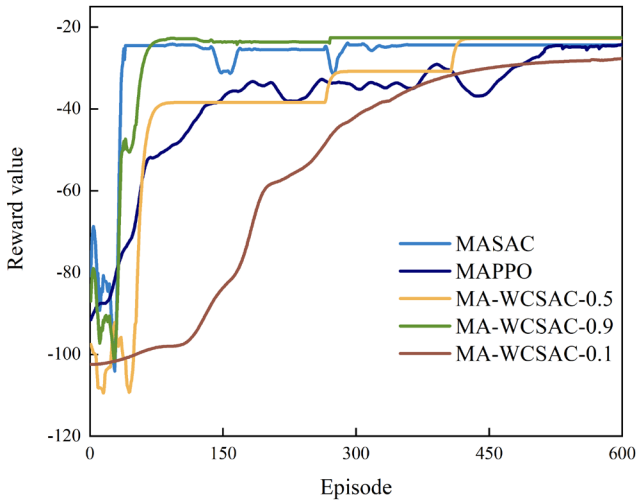


Fig. 5. Reward convergence curves for different algorithms.

As shown in Fig. 5, comparing the three cases with different values of ϖ_g, MA-WCSAC-0.1 avoids risky regions by taking a detoured but safer path, resulting in the lowest final reward among the three. MA-WCSAC-0.5 and MA-WCSAC-0.9 both tend to achieve higher reward returns overall. In particular, since MA-WCSAC-0.9 exhibits low risk aversion, its behavior is closer to that of MASAC. Overall, the degree of risk aversion in the learned policy is consistent with the theoretical analysis presented earlier: the smaller the value of ϖ_g, the more conservative the policy.

Although safety constraints may limit exploration, MA-WCSAC-0.5 achieves higher rewards by leveraging a distributional safety critic for improved risk assessment and a Lagrangian mechanism that adaptively regulates penalties. This enables stable and efficient exploration within safe boundaries, resulting in higher cumulative rewards than MASAC.

B. Distribution network reconfiguration performance

To verify the effectiveness of the proposed method in implementing DNR, this subsection analyzes three key distribution network control indicators: feeder load balance, voltage deviation and power loss.

First, the operation of CSs and OLTCs is described. The branch status changes are shown in Fig. 6, where 1 denotes a closed branch and 0 denotes an open branch. The CS switching scheme proposed by the agents is as follows: after control, branch (8, 54) is opened and branch (34, 56) is closed; branch (83, 88) is opened and branch (66, 67) is closed; branch (81, 96) is opened and branch (65, 80) is closed. In this case, each OLTC has 11 tap positions ranging from {-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5}. OLTC1 and OLTC5 adjust their tap positions to 1 and 2 respectively. The remaining OLTCs remain unchanged.

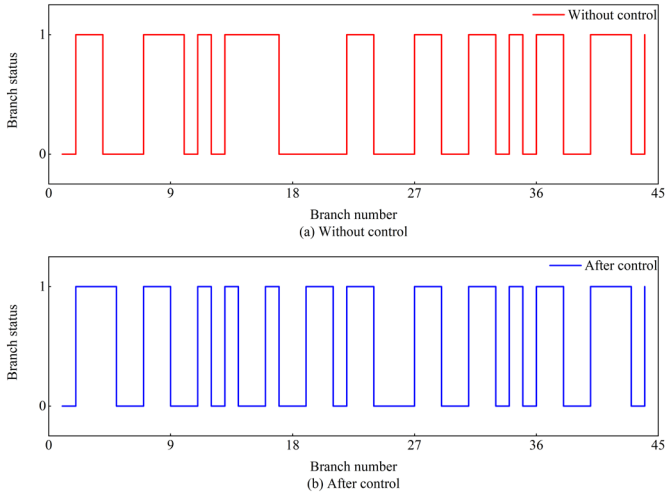


Fig. 6. Switch state before and after control.

Fig. 7 illustrates the changes in load balance across six 35kV feeders (nodes 4 to 9) before and after DNR. Compared with Fig. 7(a), the light beige area in Fig. 7(b) decreases. Prior to DNR, feeder node 6 experienced significant load imbalance during the peak hours from 12:00 to 17:00. After DNR, the load imbalance of feeder node 6 decreased by approximately 30%, indicating a notable improvement in load balance on the 35kV side. Together with the legend, the results demonstrate that the proposed method effectively achieves a more balanced load distribution among the distribution feeders.

Analyze the specific mechanism by which the agents alleviate feeder imbalance based on the actual DNR actions. By opening switch (34, 56) and closing switch (8, 54), the load (i.e., nodes 54 to 57) originally supplied by feeder node 6 is transferred to feeder node 8. Since node 8 was previously lightly loaded, it effectively accommodated the additional transferred load.

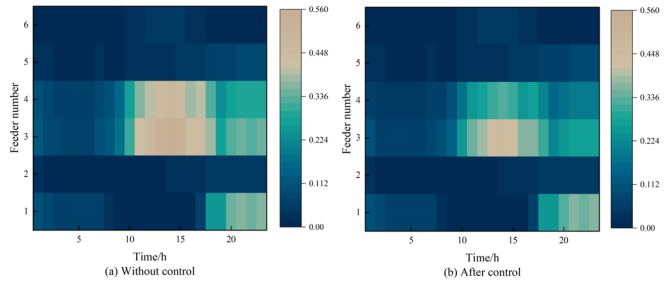


Fig. 7. Feeder load imbalance degree before and after control

Further analysis of the voltage control effect before and after DNR is presented in Fig. 8, which illustrates the voltage profiles of all nodes across different time steps. As shown in Fig. 8(a), although nodes 67 to 68 and 83 did not violate the lower voltage limit, their voltages approached the threshold of 0.95 p.u. Meanwhile, due to heavy load demand, the voltages at nodes 271 to 273 dropped below 0.95 p.u., with node 273 reaching as low as 0.9485 p.u. In contrast, Fig. 8(b) shows that after control, all node voltages were maintained above 0.962 p.u., indicating improved power quality.

The mechanism by which the agents reduce voltage deviation can be analyzed in conjunction with its specific actions. By opening branch (66, 67), closing branch (83, 88) and increasing the tap position of OLTC1 to 1, the voltages at nodes 67 to 68 and 83 were improved, as these nodes were switched to receive power supply from node 88. Similarly, by raising the tap position of OLTC5 to 2, the voltages at nodes 270 to 273 were increased. If branch (258, 259) were opened and branch (262, 168) closed, too many nodes would be transferred to node 168, resulting in excessive supply pressure on feeder node 160. Therefore, increasing the OLTC5 tap position is a better choice in this case.

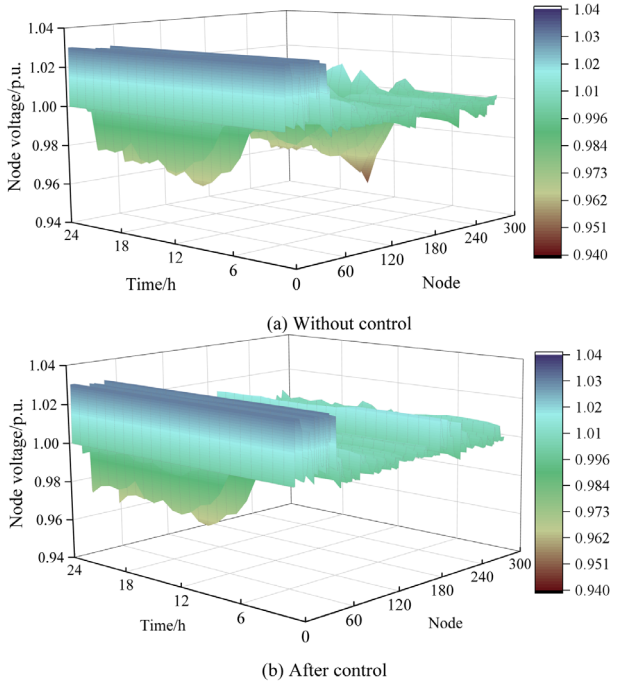


Fig. 8. Node voltage comparison before and after control.

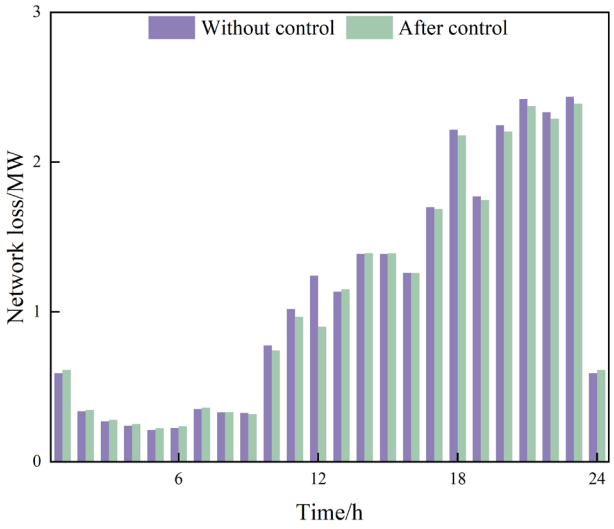


Fig. 9. Time-varying network losses.

Further analysis of the loss reduction effect before and after DNR is conducted. As shown in Fig. 9, network losses decrease across nearly all time periods following DNR and OLTC operations. According to the analyses in Fig. 7 and 8, the improved load balance among feeders leads to a more uniform load distribution, while the reduced voltage deviation enhances system stability and mitigates inter-branch current flows caused by voltage fluctuations. These factors collectively contribute to the reduction in network losses.

C. *Comparison of different algorithms*

To verify the superiority of the proposed method in DNR and its ability to ensure safe power supply under extreme conditions, this section compares its control performance with that of the MASAC and MAPPO algorithms. This subsection sets up three scenarios for comparative analysis:

Scenario 1: Baseline case.

Scenario 2: Surge in load demand and shortage of solar resources. The PV output is scaled by 0.8 and the load demand is scaled by 1.2 based on the baseline case.

Scenario 3: Reduced load demand and abundant solar resources. The PV output is scaled by 1.2 and the load demand is scaled by 0.8 based on the baseline case.

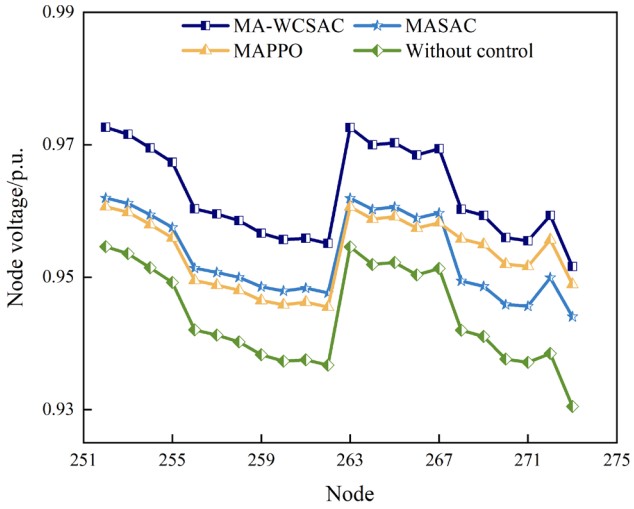


Fig. 10. Network loss comparison of different algorithms under Scenario 3.

Fig. 10 illustrates the voltage distribution of nodes 252 to 273, which are connected to the root node 232 at the peak load time of 13:00 under Scenario 2. As previously analyzed, before control, due to the large number of load nodes and the peak demand period, root node 232 experienced significant supply pressure, causing the voltage of multiple nodes to drop below 0.95 p.u. After the agent actions, the voltage drop issue at root node 232 and its connected nodes was alleviated, with all node voltages including that of the terminal node 273, maintained above 0.95 p.u.. Compared to the MASAC and MAPPO algorithms (which have partially mitigated the voltage drop issue), the proposed method achieves more significant improvements in the voltage profile around root node 232. This indicates that the introduction of hard safety constraints and the distributional safety critic network enhances the agents' power supply capability and robust control under high-risk conditions.

TABLE III presents the performance of different algorithms over one operation cycle in terms of each optimization metric (as defined in equation (22)) and the overall objective function. In the table, action cost comprehensively accounts for both OLTC and CS operation costs. In addition to the DRL-based methods mentioned in Section V.A, TABLE III also introduces

a traditional model-based methodology, namely the MISOCP approach. Considering that the MISOCP requires a long computation time in this case study—exceeding 12 hours—the results presented in this paper correspond to the solution obtained at the 12-hour cutoff. For the voltage-related metrics, in addition to the voltage deviation, voltage violation is also evaluated. Specifically, the violation value $VV_{j,t}$ is first calculated according to equation (48) and then the mean (Mean) and standard deviation (Std.) of $VV_{j,t}$ are computed.

$$VV_{j,t} = \begin{cases} 0.95 - U_{j,t}, U_{j,t} < 0.95 \\ U_{j,t} - 1.05, U_{j,t} > 1.05 \\ 0, \text{else} \end{cases} \tag{48}$$

The proposed algorithm improves extreme-risk evaluation and effectively ensures control safety. In the severely overloaded Scenario 2, its voltage violation rate is significantly lower than that of the other two algorithms, demonstrating strong robustness under extreme conditions. In Scenario 3, it successfully prevents voltage violations caused by excessive PV generation. Overall, across the three scenarios, MA-WCSAC outperforms MASAC and MAPPO in most optimization metrics, indicating its ability to avoid local optima associated with single-metric optimization. Although the MISOCP method often achieves more optimal solutions than DRL-based approaches, its excessive computation time limits practical applicability. A comparison of computation times is provided in Table IV in Section V.D.

TABLE III
OPTIMIZATION RESULTS OF DIFFERENT METRICS ACROSS DIFFERENT ALGORITHMS UNDER DIFFERENT SCENARIOS.

Method	Voltage/p.u.			Network loss/MW	Load balance indicator	Action cost	Total objective
	Deviation	Mean	Std.				
Scenario 1							
Proposed	14.473	0	--	26.334	14.110	3.9	22.540
MASAC	18.565	$1.586e\times10^{-4}$	1.746×10^{-5}	27.252	16.013	2.9	27.499
MAPPO	17.739	9.541×10^{-5}	6.298×10^{-6}	26.964	17.309	4.7	25.298
MISOCP	14.894	0	--	25.876	13.992	4.5	22.085
Scenario 2							
Proposed	21.006	8.205×10^{-4}	3.989×10^{-4}	33.778	14.832	5.6	29.290
MASAC	38.805	8.288×10^{-3}	5.814×10^{-3}	46.742	18.015	3.8	43.667
MAPPO	29.262	5.497×10^{-3}	3.028×10^{-3}	40.248	17.662	4.8	34.489
MISOCP	20.549	8.205×10^{-4}	3.989×10^{-4}	29.136	14.045	5.3	23.085
Scenario 3							
Proposed	12.382	0	--	21.186	15.477	2.7	19.291
MASAC	15.327	0	--	21.854	14.410	2.9	20.775
MAPPO	17.121	1.706×10^{-5}	6.106×10^{-8}	23.234	17.684	1.9	22.807
MISOCP	11.974	0	--	19.942	13.092	4.0	18.357

D. Advantages of cloud-edge collaboration with physically interpretable agent partitioning

This section further elaborates on the advantages of the proposed cloud-edge collaboration method with physically interpretable agent partitioning. To validate its effectiveness, three comparative schemes are designed: (i) the proposed method (M1), where agents are partitioned according to feeder groups and actual physical connections under a cloud-edge collaborative framework; (ii) a non-collaborative approach (M2), where a single agent controls the entire system without cloud-edge collaboration; (iii) a fine-grained partitioning method (M3), where each CS and OLTC is assigned an individual agent, resulting in a larger number of small-scale agents. The modeling approach can be referred to in [4].

From a computational perspective, M2 and M3 exhibits combinatorial complexity with respect to the number of switches $O(2^n)$, whereas the proposed partitioned framework M1 reduces effective decision dimensionality to $O(n^2)$.

TABLE IV
RESULTS OF DIFFERENT METRICS UNDER DIFFERENT CONTROL METHOD.

Method	Voltage/p.u.			Network loss/MW	Load balance indicator	Action cost	Total objective	Test time/s
	Deviation	Mean	Std.					
M1	14.473	0	--	26.334	14.110	3.9	22.540	16.83
M2	18.174	1.225×10^{-4}	1.042×10^{-5}	26.805	15.664	1.0	22.807	19.32
M3	14.924	0	--	26.415	13.329	6.3	23.124	45.49
MISOCP	14.894	0	--	25.876	14.992	5.5	23.085	>12h
No control	19.582	2.692×10^{-4}	5.032×10^{-5}	27.471	16.930	0	24.749	\

TABLE IV presents a comparative evaluation of different control strategies in terms of performance metrics and test-set computation time under the baseline scenario (Scenario 1). The proposed cloud–edge coordination framework adopts a hierarchical structure, in which partitioned edge agents handle local operational characteristics, while the cloud layer performs global coordination to satisfy network-wide coupling constraints. This design allows each edge agent to focus on its most relevant decision variables and constraints, while preserving global consistency.

Moreover, the agent partitioning strategy provides clear physical interpretability and achieves a balanced trade-off between centralized and decentralized control. It overcomes the limitations of M2, where excessive centralization forces a single agent to manage an excessively large action space, leading to severe dimensionality issues that hinder learning efficiency, scalability and the representation of distinct operational characteristics across distribution network layers.

These results indicate that, even though DNR operations are not performed frequently, a decentralized design remains valuable for improving scalability and adaptability in large-scale distribution systems.

On the other hand, the proposed approach also avoids the redundant overhead and coordination complexity of M3, in which excessive decentralization results in a large number of agents and rapidly increasing computational costs as the network scales, thereby complicating training. Consequently, the proposed method achieves a favorable balance between centralized and decentralized designs. In contrast, the MISOCP approach requires more than 12 hours of computation time, making it impractical for real-time engineering applications.

VI. Conclusion

This paper proposes a cloud–edge collaborative DNR framework based on MA-WCSAC safe reinforcement learning. Instead of relying on purely deterministic optimization, the proposed method integrates feeder-group-based physically interpretable agent partitioning, Lagrangian hard safety constraints, and distributional tail-risk modeling into a unified learning framework. Case studies demonstrate three main findings. First, the proposed method enhances robustness under extreme conditions by reducing voltage violations and ensuring secure operation in heavy-load and high-renewable scenarios. Second, the distributional safety critic strengthens risk awareness compared with baseline methods, resulting in more stable control performance. Third, feeder-group-based agent partitioning under cloud–edge collaboration achieves a better balance between scalability and global coordination than centralized or overly decentralized schemes.

From an engineering perspective, the proposed framework provides a scalable and risk-aware solution for large-scale active distribution networks with high penetration of distributed energy resources. It is particularly suitable for real-time or near-real-time operation scenarios supported by cloud–edge coordinated control platforms.

Future work may focus on designing collaborative decision-making mechanisms under limited information exchange, incorporating communication constraints among agents and validating the approach in practical utility-level distribution systems with real operational data.

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